Lab 28 Worksheet — Network Fault Diagnosis — Command Line to Packet Level

Name: ____ **Date:** __

Exam objective: Diagnose network faults by working the layers in order, from physical connectivity through addressing and DNS to application behaviour (Core 1 objective 5.7).

Record as you go

Step	What you did	What you observed
1	Build the diagnostic sequence bottom-up: physical link, IP addressing, default gateway, DNS resolution, then the application itself. Testing out of order produces misleading results.	
2	Open the Killercoda Ubuntu playground and install the diagnostic tool set.	
3	Layer 1 — confirm the interface is physically up and has carrier, since every layer above depends on it.	
4	Layer 2 and 3 — record the interface addresses and identify whether the address is valid, or an APIPA 169.254 address meaning DHCP failed.	
5	Verify the address and mask in IP Calculator at https://alfredang.github.io/ipcalculator/ and confirm the host sits inside its own subnet's usable range.	
6	Layer 3 — confirm a default route exists, then test reachability to the gateway. No default route means no traffic can leave the subnet.	
7	Test reachability beyond the gateway by IP address, which isolates routing from DNS entirely.	
8	Layer 7 — test DNS resolution separately,	

Step	What you did	What you observed
	because a machine that pings an IP but not a name has a DNS fault and nothing else.	
9	Trace the path to identify where latency is introduced or where the path stops.	
10	Check active connections and listening sockets to confirm the application layer is actually working.	
11	Capture traffic and analyse it in PCAP Analyzer when the commands are inconclusive, since the packets show what the tools only summarise.	
12	Build the symptom map for APIPA address, no default gateway, pings IP but not name, high latency, jitter degrading VoIP, port flapping, intermittent wireless and limited connectivity — giving the layer, the test and the fix for each.	

Verification

Success criterion: Every command in the sequence runs successfully on the playground, the sequence tests layers strictly bottom-up, IP Calculator confirms the host address sits within its usable range, and the symptom map assigns a layer and a specific test to all eight faults.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
